1 FIG-216 — Three-model MLIP static force response across candidate stages

Group: MD-FROM-SCRATCH-FAILURE-ATTRIBUTION **Evidence chain:** manifested **Status:** paired static MLIP force response; DFT correctness pending

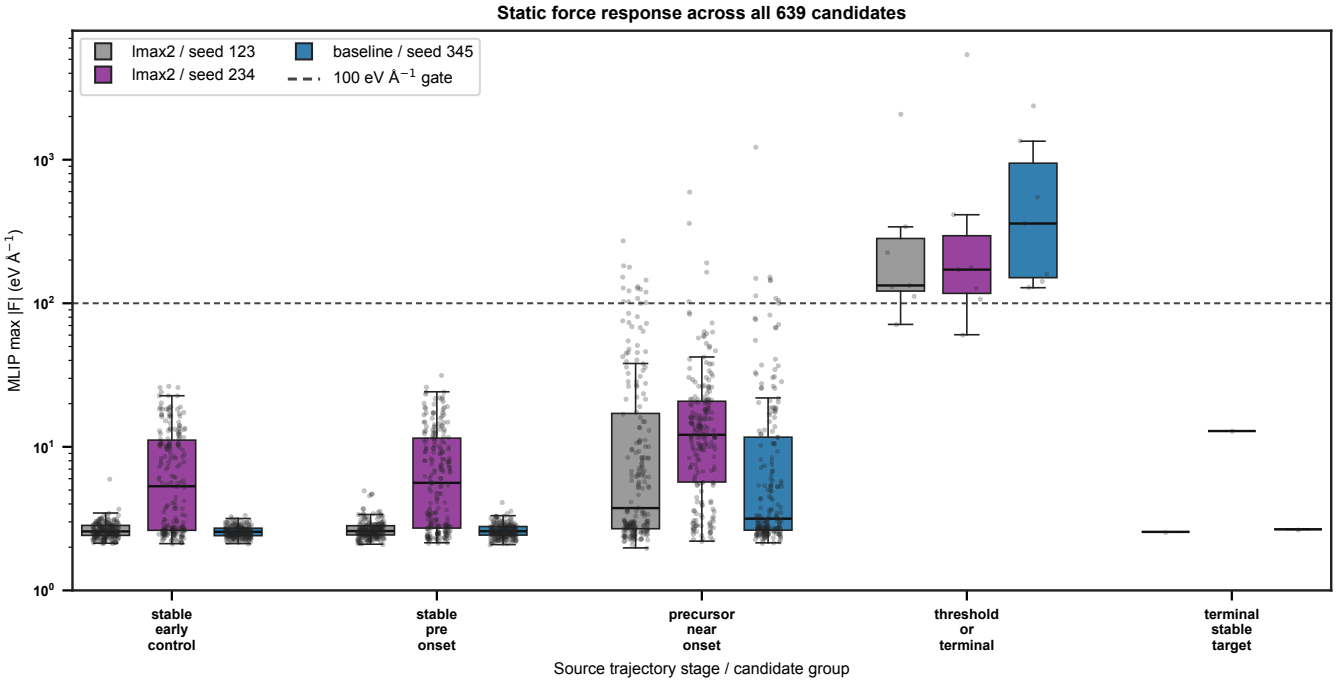


Figure 1: Paired static maximum-force responses for the same 639 fully periodic 2050-atom candidate structures evaluated with three registered checkpoints. Source-stage labels are retained from the audited trajectories; the logarithmic axis displays stable, precursor, and threshold tails together. The 100 eV Å^{-1} line is the MD instability gate, not a DFT correctness criterion.

Results text draft. The campaign contains 1,917 finite model–candidate evaluations with no runtime failures. Median maximum force is approximately 2.6–5.6 eV Å^{-1} in the stable controls, rises to 3.2–12.1 eV Å^{-1} in the precursor group, and reaches 133, 172, and 359 eV Å^{-1} for the three checkpoints in the threshold-or-terminal group. These are static responses on saved structures, not new MD onset times. **Discussion and boundary.** The stage-wise increase and the larger threshold response of the seed-234 lmax-2 and baseline checkpoints are consistent with model-dependent force amplification in the extrapolative region. This supports a diagnostic hypothesis about model sensitivity, but does not identify the DFT-correct force, prove that the lmax reduction caused the runaway, or replace the stochastic MD evidence. Energies are retained for provenance and are not absolutely ranked across model families. **Article placement.** Proposed placement: Supplementary Information / from-scratch instability and three-model static force audit; not a standalone DFT-validation or transport figure. **Reviewer linkage:** overall status PENDING.

- **REV-C9 (PENDING)**: Assess whether the all-candidate paired MLIP response supports the proposed instability follow-up. *Completed:* 639 source candidates, three checkpoint hashes, and finite force outputs are covered. *Still required:* independent reference calculations before a correctness or causal claim.

Evidence pointers.

- Results and discussion: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/figures/MLIP_RESULTS_DISCUSSION.md
- Processed result table and manifest: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/model_stage_summary.csv; /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/analysis_manifest.json

- Plot script and Plot Atlas source: /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/make_mlip_analysis.py; /Users/alina/Documents/Codex/plot_library
- Asset SHA256: 454b4ef234c40579fbd4a6a61b9c551f7056047950ac9d537cde2069

Machine-collected evidence candidates (generated from the analysis manifest; do not edit this file).

- **Analysis manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/results/analysis_manifest.json
- **Model registry:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/mlip_candidate_campaign_20261013/MODEL_REGISTRY.tsv
- **Plot output manifest:** /Users/alina/Documents/ChatGPT/Wiley-MigrationBench/analysis/from_scratch_failure_attribution_20261011/mlip_candidate_campaign_20261013/figures/figure_and_text_manifest.json

Audit gap / next action: Compare the MLIP-selected local periodic reconstructions with converged QE calculations and retain full-cell parity, pseudopotential, and convergence checks before promoting the force comparison.